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INDIVIDUAL LABORATORY REPORT

Lab 3: 3D Trajectory Following

Reference Generation and Experimental Validation

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Experimental Validation of a Geometric Quadrotor Controller for Three-Dimensional Trajectory Following

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***Abstract*—This report examines the reference generation and experimental validation of a quadrotor geometric controller developed jointly by Group 6 for MIT VNAV Lab 3. My responsibility is Role D: reference-trajectory analysis, test comparison, telemetry evaluation and presentation of the recorded flight. The official Unity/TESSE simulator was run through Box64 0.3.6 in an ARM64 ROS Noetic container. The reference is a horizontal circle of radius 5 m, at a target height of 2 m, with a nominal period of 12.566 s. The final record contains 2,752 state samples over 55.02 s. Using the team's existing two-period exclusion, the evaluation window contains 2.376 observed turns, a three-dimensional position RMS error of 0.3063 m and a maximum error of 0.5845 m. The mean height error is –0.2532 m; vertical error contributes 72.1% of the squared position error. The analysis therefore distinguishes successful repeated circular motion from precise three-dimensional tracking, and identifies the limitations of visual inspection, gain comparisons and asynchronous reference sampling.**

I. Introduction and Personal Scope

The experiment connects a desired trajectory to the forces and moments required to move a simulated quadrotor. Following the Lab 3 task, the system uses ROS messages for the desired and measured states, geometric feedback for position and attitude, and four signed rotor-speed commands for the official TESSE simulator [1], [2]. Although the exercise is called *3D trajectory following*, this particular test follows a planar circle at constant height. It exercises three-dimensional position and attitude control; it is not a demonstration of an arbitrary spatial path or obstacle-avoidance planning.

This is my individual submission for a jointly completed experiment. My assigned contribution is **Role D: Reference Trajectory and Experimental Validation**, within the group's equal-contribution arrangement (25% per member). I focus on the analytical reference, the organisation of hover and circular tests, the interpretation of the recorded telemetry, and the figures used to assess tracking. The controller and runtime are shared group work, not an independently developed system attributed solely to me.

The main question for my part is: *what evidence is sufficient to say that the vehicle follows the requested trajectory?* A moving vehicle in RViz is necessary visual evidence, but it does not quantify height error, timing mismatch or the effect of the initial transient. I therefore use the retained CSV files and validation records alongside the images [3]. The quaternion-conversion questions (Deliverable 1) and rotor-layout/drag-modelling derivations (Deliverable 2) are supplied as separate individual submissions; this report concentrates on the shared flight experiment and my validation role.

II. System and Geometric Control

A. From the reference to measured motion

The final experiment uses the unmodified official Lab 3 Unity/TESSE player. The team's validation record identifies ARM64 Docker, ROS Noetic, Box64 0.3.6 and a self-contained workspace at `/tmp/lab3_unity_ws`. The Python bridge uses Python 3 and loads `cv2` before `cv_bridge`; this compatibility change concerns module initialisation, not the control equations [3].

The trajectory publisher supplies the desired state. The controller compares it with the most recently received measured state, computes the requested wrench, and publishes rotor commands. The quadrotor control interface sends these commands to TESSE; the bridge returns the simulated state. A separate monitoring script records states together with the latest received reference and command, and publishes the actual and desired paths for RViz [2], [3].

B. Coordinate conventions and geometric feedback

Let $\boldsymbol{x}, \boldsymbol{v} \in \mathbb{R}^3$ denote position and linear velocity in the world frame, and let $R \in SO(3)$ rotate body-frame vectors into world coordinates. The Lab 3 convention uses an upward z axis, with $\boldsymbol{e}_3 = (0,0,1)^\mathsf{T}$. For this TESSE interface, the received angular velocity is treated as a world-frame vector and transformed as $\boldsymbol{\Omega} = R^\mathsf{T}\boldsymbol{\Omega}_w$ before the body-frame attitude calculation. This is the experiment's stated convention; it should not be inferred merely from the generic Odometry message name [2].

The shared controller adapts the geometric construction of Lee, Leok and McClamroch [4]. With position and velocity errors defined as actual minus desired, the requested force vector is

$$\begin{aligned} \boldsymbol{e}_x &= \boldsymbol{x} - \boldsymbol{x}_d, & \boldsymbol{e}_v &= \boldsymbol{v} - \boldsymbol{v}_d, \\ \boldsymbol{A} &= -k_x\boldsymbol{e}_x - k_v\boldsymbol{e}_v + mg\boldsymbol{e}_3 + m\boldsymbol{a}_d. \end{aligned} \tag{1}$$

The gravity term has a positive sign here because the thrust must counter downward gravity. The feedforward term $m\boldsymbol{a}_d$ supplies the acceleration demanded by the moving reference rather than leaving all of that demand to feedback.

Table I
INTERFACES RELEVANT TO REFERENCE GENERATION AND EVALUATION.

Topic	Message type	Role in the experiment
`/desired_state`	`trajectory_msgs/MultiDOFJointTrajectoryPoint`	Desired pose, velocity, acceleration and elapsed reference time.
`/current_state`	`nav_msgs/Odometry`	Measured position, orientation and velocities from TESSE.
`/rotor_speed_cmds`	`mav_msgs/Actuators`	Four signed angular velocities in rad/s.
`/clock`	`rosgraph_msgs/Clock`	Simulation time used by the ROS reference and timers.

For a nonzero requested force and a nonparallel heading construction,

$$\begin{aligned} \boldsymbol{b}_{3d} &= \frac{\boldsymbol{A}}{\|\boldsymbol{A}\|}, & \boldsymbol{b}_{1c} &= (\cos\psi_d, \sin\psi_d, 0)^{\mathsf{T}}, \\ \boldsymbol{b}_{2d} &= \frac{\boldsymbol{b}_{3d}\times\boldsymbol{b}_{1c}}{\|\boldsymbol{b}_{3d}\times\boldsymbol{b}_{1c}\|}, & R_d &= [\,\boldsymbol{b}_{2d}\times\boldsymbol{b}_{3d} \quad \boldsymbol{b}_{2d} \quad \boldsymbol{b}_{3d}\,]. \end{aligned} \tag{2}$$

The implementation supplies fallback directions for the degenerate cases. Its attitude error, thrust and moment are

$$\begin{aligned} \boldsymbol{e}_R &= \tfrac{1}{2}(R_d^{\mathsf{T}}R - R^{\mathsf{T}}R_d)^{\vee}, \\ f &= \boldsymbol{A}^{\mathsf{T}}R\boldsymbol{e}_3, \\ \boldsymbol{M} &= -k_R\boldsymbol{e}_R - k_\Omega\boldsymbol{\Omega} + \boldsymbol{\Omega}\times J\boldsymbol{\Omega}. \end{aligned} \tag{3}$$

Here $(\cdot)^{\vee}$ converts a skew-symmetric matrix into its corresponding vector. The desired angular-rate feedforward terms are omitted under the simplification allowed by the exercise. Thus the implemented law is not the complete feedforward law in the reference paper [2], [4].

The controller retains $m = 1\,\mathrm{kg}$, $d = 0.3\,\mathrm{m}$, $c_f = 10^{-3}$, $c_d = 10^{-5}$ and $J = I_3$ from the skeleton model. The validation record notes that Unity's internal inertia differs. With $a = d/\sqrt{2}$ and $s_i = \omega_i|\omega_i|$, its simulator-specific allocation is

$$B_{\mathrm{sim}} = \begin{bmatrix} c_f & c_f & c_f & c_f \\ ac_f & ac_f & -ac_f & -ac_f \\ -ac_f & ac_f & ac_f & -ac_f \\ c_d & -c_d & c_d & -c_d \end{bmatrix}. \tag{4}$$

With $\boldsymbol{w} = (f, M_x, M_y, M_z)^{\mathsf{T}}$ and $\boldsymbol{s} = (s_1, s_2, s_3, s_4)^{\mathsf{T}}$, the command satisfies $\boldsymbol{w} = B_{\mathrm{sim}}\boldsymbol{s}$ and $\omega_i = \mathrm{sgn}(s_i)\sqrt{|s_i|}$. The simulator uses a rotor numbering convention different from the separate written Deliverable 2 exercise. The signed-square convention preserves the simulator's reversible rotor commands; it is not a hardware-safety recommendation for a physical quadrotor [2].

III. REFERENCE-TRAJECTORY DESIGN

A. Analytical circle and its derivatives

The final tests preserve the supplied circle rather than changing its radius or speed to make tracking easier. With radius $r = 5\,\mathrm{m}$, time scale $\tau = 2\,\mathrm{s}$ and target height $h = 2\,\mathrm{m}$, the reference is

$$\begin{aligned} \boldsymbol{x}_d(t) &= \begin{bmatrix} r\sin(t/\tau) \\ r\cos(t/\tau) \\ h \end{bmatrix}, \\ \boldsymbol{v}_d(t) &= \begin{bmatrix} (r/\tau)\cos(t/\tau) \\ -(r/\tau)\sin(t/\tau) \\ 0 \end{bmatrix}, \\ \boldsymbol{a}_d(t) &= \begin{bmatrix} -(r/\tau^2)\sin(t/\tau) \\ -(r/\tau^2)\cos(t/\tau) \\ 0 \end{bmatrix}, \\ \psi_d(t) &= -t/\tau, \qquad \dot{\psi}_d = -1/\tau. \end{aligned} \tag{5}$$

These expressions match the shared trajectory publisher [2]. Direct differentiation gives the velocity and acceleration; it avoids estimating derivatives from noisy or irregularly received positions.

Three checks are useful before assessing the controller:

$$\begin{aligned} \|\boldsymbol{v}_d\| &= r/\tau = 2.5\,\mathrm{m/s}, \\ \|\boldsymbol{a}_d\| &= r/\tau^2 = 1.25\,\mathrm{m/s^2}, \\ T &= 2\pi\tau = 4\pi \approx 12.566\,\mathrm{s}. \end{aligned} \tag{6}$$

Constant speed does not mean zero acceleration: the velocity direction changes continuously. Also, $\boldsymbol{v}_d^{\mathsf{T}}\boldsymbol{a}_d = 0$, which is consistent with centripetal acceleration. At $t = 0$, the reference starts at $(0, 5, 2)^{\mathsf{T}}$ with velocity $(2.5, 0, 0)^{\mathsf{T}}$. From above, it initially moves towards positive x and proceeds clockwise. The horizontal velocity direction is $(\cos(t/\tau), -\sin(t/\tau))^{\mathsf{T}}$, so $\psi_d = -t/\tau$ makes the requested heading tangent to the circle.

The ROS message carries one transform, one velocity and one acceleration. The following excerpt shows the pose and elapsed-time fields; t is simulation time, not the computer's wall-clock time.

```
const double phase = t / time_scale;
p.x = radius * std::sin(phase);
p.y = radius * std::cos(phase);
p.z = height;
q.setRPY(0.0, 0.0, -phase);
msg.time_from_start = ros::Duration(t);
```

The velocity and acceleration fields use the analytical derivatives in Equation (5); their linear z and angular-acceleration components are zero. The desired yaw rate is stored as $-1/\tau$ even though the simplified controller does not use full desired attitude-rate feedforward.

B. Hover, start-up and the meaning of time

The static mode requests $(0,0,2)^{\mathsf{T}}$, zero velocity and acceleration, and yaw $\pi/4$. This removes the circular acceleration demand and gives a simpler preliminary test. The archived official hover run lasts 25.04 s; it is separate from the final circular record [3].

The publisher waits for valid ROS time, uses `/clock`, and is configured for 100 Hz. The control timer is configured for 200 Hz in the final Unity gains. These are requested rates, not measurements of delivered update rates. The final CSV starts at an actual position of approximately $(1.884, 4.118, 0.866)$ m while the desired position is $(0,5,2)$ m. Therefore, the recorded run is not a perfect-on-reference initialisation. A nonzero start-up error is part of the evidence and must remain visible.

IV. Experimental Protocol and Metrics

A. Records and parameter comparisons

I organise the retained official-simulator records as shown in Table II. No additional flight is claimed in this report: the analysis uses the archived runs. In particular, the file named `unity_circle_tuned.csv` is the higher-gain trial that was *rejected*, not the final selected result [3].

The first two circular tests share the same 20 s threshold. The final recorded run instead uses the team's existing two-period threshold, $t_s = 2T = 8\pi \approx 25.133$ s. Its first included reference timestamp is 25.18 s, because the sampled record does not contain a state/reference pair at every possible reference time. I retain these conventions rather than silently giving every run the same label of "steady state". Here, *steady window* is the documented evaluation interval, not a formal settling-time result.

The final CSV contains 2,752 rows. Its state timestamps increase at approximately 0.02 s intervals, but 1,832 adjacent pairs have the same reference-time value. They are not duplicate vehicle states: the monitor pairs each new state with the most recently received reference. Removing these rows would change the weighting of the reported metric. The final calculation therefore keeps all rows selected by the established reference-time threshold.

B. Metrics and their interpretation

Let $\boldsymbol{e}_i = \boldsymbol{x}_i - \boldsymbol{x}_{d,i}$ be the error relative to the reference stored in row i, and let $\mathscr{S}$ be the selected evaluation rows. The reported metrics are

$$
\begin{aligned}
E_{\mathrm{RMS}} &= \sqrt{\frac{1}{|\mathscr{S}|}\sum_{i\in\mathscr{S}} \|\boldsymbol{e}_i\|^2}, \\
E_{\max} &= \max_{i\in\mathscr{S}} \|\boldsymbol{e}_i\|, \\
E_r &= \sqrt{\frac{1}{|\mathscr{S}|}\sum_{i\in\mathscr{S}} \left(\sqrt{x_i^2+y_i^2} - r\right)^2}, \\
E_j &= \sqrt{\frac{1}{|\mathscr{S}|}\sum_{i\in\mathscr{S}} e_{j,i}^2}, \\
&\qquad j \in \{x,y,z\}.
\end{aligned}
\tag{7}
$$

RMS error describes the overall magnitude over the window; the maximum records the worst sample. Radial error measures distance from the circular locus, whereas position error also penalises being at the wrong phase of that circle. A vehicle can therefore have a small radial error while remaining behind the desired position.

To verify actual motion rather than count requested periods, the observed turn count uses

$$
\theta_i = \mathrm{unwrap}(\mathrm{atan2}(x_i, y_i)), \qquad N_{\mathrm{turns}} = \frac{\theta_{\mathrm{last}} - \theta_{\mathrm{first}}}{2\pi}. \tag{8}
$$

The order $\mathrm{atan2}(x,y)$ matches the reference $x = r\sin(t/\tau)$, $y = r\cos(t/\tau)$. Unwrapping prevents the branch cut from being counted as a reversed rotation. This calculation establishes angular progress around the circle, not a general stability guarantee.

The component decomposition is useful because

$$
E_{\mathrm{RMS}}^2 = E_x^2 + E_y^2 + E_z^2, \qquad \eta_z = \frac{E_z^2}{E_{\mathrm{RMS}}^2}. \tag{9}
$$

The quantity η_z attributes the squared error to its vertical component. It is not an uncertainty estimate. All statistics remain conditional on the recorded run and the chosen sampling convention.

C. Recording and timing limits

The full RViz recording is 57.125 s long at 1280×800 pixels and 8 frames per second. The submitted excerpt starts at wall-video time 30 s and lasts 27.125 s, without accelerated playback. Video duration, reference time and ROS bag duration are different quantities and are not interchangeable [3].

The saved bag inventory reports 2,821 state messages, 2,614 desired-state messages and 2,706 rotor-command messages over a 56.4 s recorded simulation span. Dividing these counts by that span gives approximately 50.0, 46.3 and 48.0 messages per recorded simulation second, respectively. These are coarse span averages, not measurements of jitter or proof of a uniform control interval. They explain why a 200 Hz timer setting should not be reported as a measured 200 Hz command stream. The original bag itself is not included in the supplied archive; these counts come from its retained inventory.

V. Results and Evaluation

A. Complete flight and observed circular motion

Figure 1 retains the complete trajectory and separates the first two nominal periods from the final evaluation window. The vehicle eventually follows repeated circles, but the initial position error is much larger than the steady-window error. Recalculation of the entire CSV gives an all-record RMS error of 3.1934 m, with a peak of 9.5700 m at reference time 7.06 s. These whole-record values describe acquisition and tracking together; they must not be replaced by the smaller steady-window result without stating the exclusion.

Within the final window, the vehicle advances by 2.3756 observed turns. The original metrics file and an independent recalculation from the supplied CSV agree on $E_{\mathrm{RMS}} =$

Table II
OFFICIAL TESSE TESTS AND THEIR RECORDED EVALUATION WINDOWS. GAINS ARE ORDERED AS $(k_x, k_v, k_R, k_\Omega)$.

Record	Gains	Duration	Window used for the stored summary
Static hover	$(2, 2.4, 1.8, 0.7)$	25.04 s	Reference time $t \geq 12.5$ s; 630 samples.
Initial circle	$(2, 2.4, 1.8, 0.7)$	40.00 s	$t \geq 20$ s; 1,001 samples.
Higher-gain circle	$(4, 3, 2.4, 0.6)$	40.18 s	$t \geq 20$ s; 1,011 samples.
Final circle	$(2, 2.4, 1.8, 0.7)$	55.02 s	$t \geq 8\pi$ s; 1,495 samples.

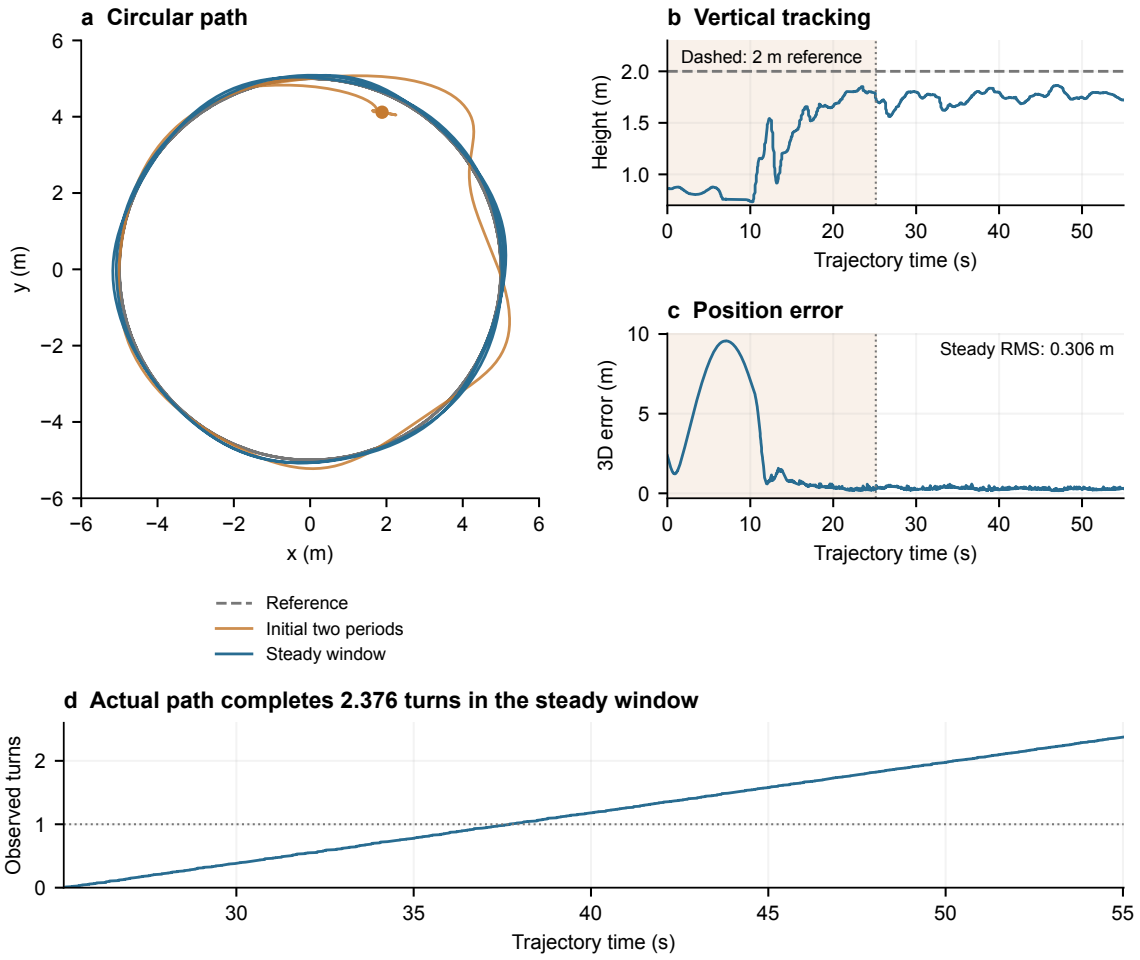


Figure 1. Official TESSE flight: horizontal path, altitude, three-dimensional position error and observed angular progress. The orange portion denotes the first two nominal periods; the blue portion denotes the final evaluation window. All measurements remain visible in the time histories. *Source: Group 6 experimental archive.*

0.306 306 m, $E_{\max} = 0.584\,545$ m and $E_r = 0.082\,926$ m [3]. This supports completion of more than one circular lap, which is the principal visual requirement. It does not establish accurate tracking throughout the start-up segment.

B. Why the plan view is not sufficient

The small radial error makes the top-view path look close to the intended circle. Figure 2 changes that interpretation: the measured path lies below the 2 m reference plane. Even the maximum height in the selected window is only 1.8638 m. Thus the height error is not just a brief overshoot in an otherwise centred trajectory.

The component RMS errors are 0.1153 m in x, 0.1135 m in y and 0.2601 m in z. Equation (9) gives $\eta_z = 0.7211$: the vertical component accounts for 72.1% of the squared three-dimensional error. Its mean error, −0.2532 m, is close in magnitude to its RMS, indicating that a negative offset contributes strongly to the vertical result. Figure 3 distinguishes that offset from the smaller oscillatory horizontal errors.

The preliminary hover record also shows a residual position error: the stored late-window RMS is 0.2452 m. Therefore, it would be premature to attribute the final height deficit solely to the centripetal acceleration of circular flight. The

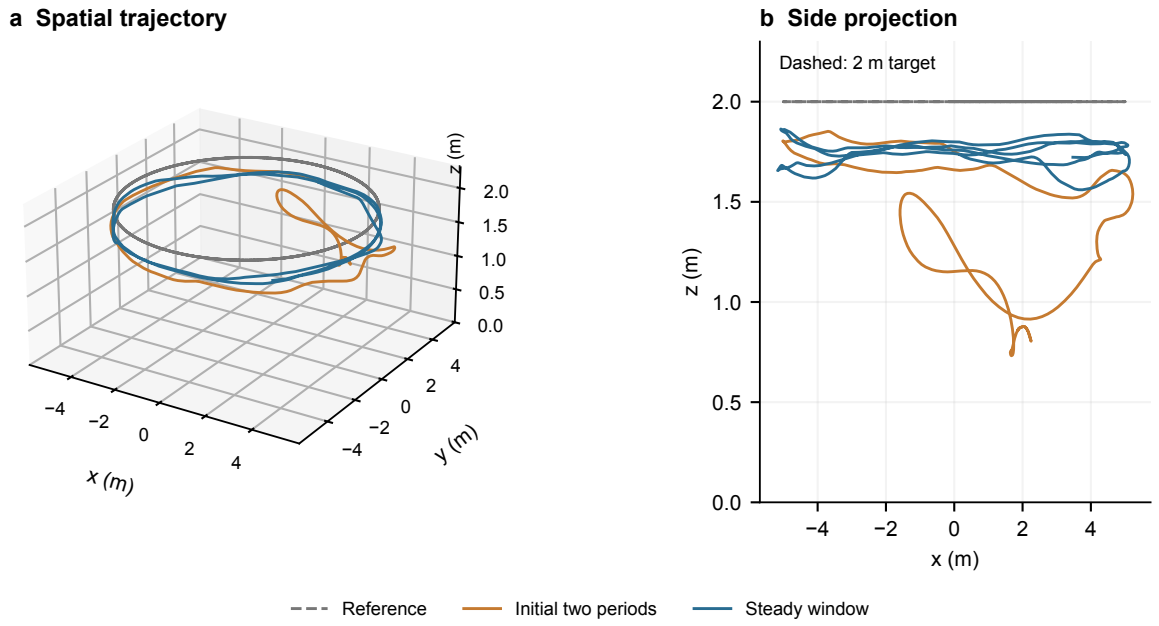


Figure 2. Three-dimensional path and side projection of the same official flight. The side view exposes a persistent vertical offset that is difficult to judge from a plan view. The reference plane remains at 2 m. *Source: Group 6 experimental archive.*

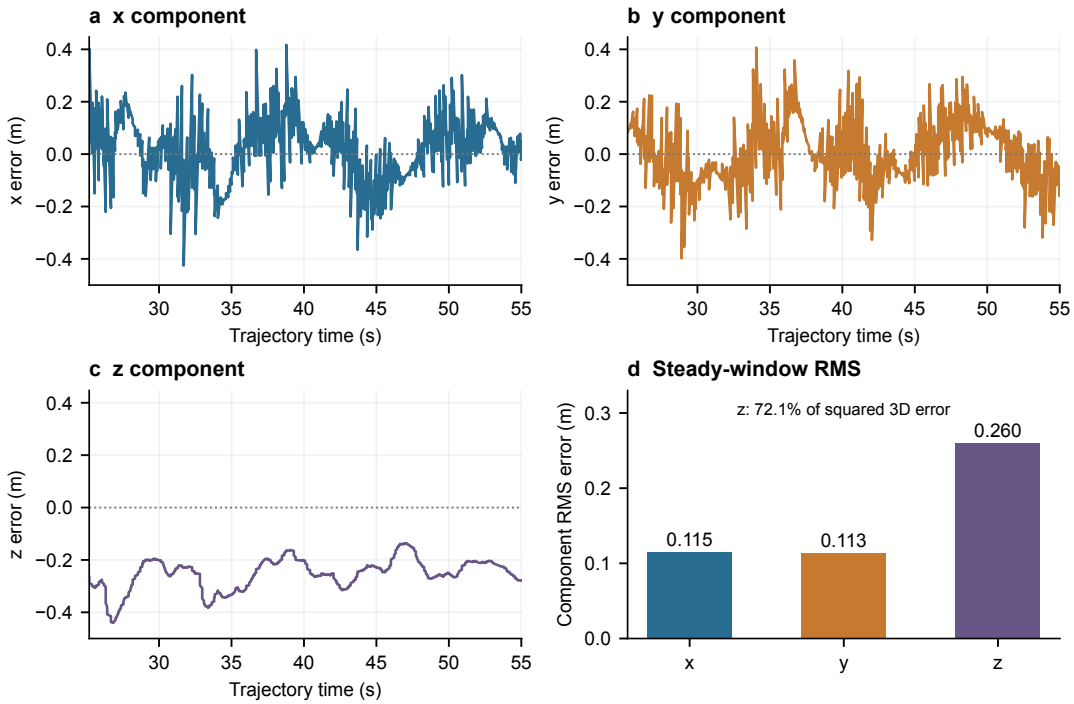


Figure 3. Error components and component RMS values in the final evaluation window. All components use the same actual-minus-desired convention. The vertical error dominates the squared-error budget, but this decomposition does not identify its physical cause. *Source: Group 6 experimental archive.*

supplied measurements support the presence of the deficit, but do not isolate thrust-model mismatch, attitude effects and timing behaviour as separate causes.

C. Parameter comparison: larger feedback is not automatically better

The initial circular test with $(2, 2.4, 1.8, 0.7)$ gives an RMS error of 0.2710 m after reference time 20 s. The alternative

Table III
FINAL CIRCULAR RECORD. POSITION ERRORS ARE COMPUTED AGAINST THE LATEST REFERENCE STORED WITH EACH STATE.

Quantity	Result
Recorded state samples	2,752
Samples in final evaluation window	1,495
First and last included reference timestamps	25.18 s–55.02 s
Observed turns in this window	2.3756
Three-dimensional position RMS error	0.3063 m
Maximum three-dimensional position error	0.5845 m
Radial RMS error	0.0829 m
Actual height range	1.5598 m–1.8638 m
Mean height error, actual minus desired	–0.2532 m
Whole-record RMS error, including start-up	3.1934 m

$(4, 3, 2.4, 0.6)$ gives 0.8970 m over the corresponding late window, so the original combination is retained for the final recording [3]. Figure 4 shows both complete histories, rather than displaying only the final aggregate values.

The final recorded run with the selected combination gives 0.3063 m, not the initial trial's 0.2710 m. The records are separate runs and their reported windows differ. I consequently do not present that difference as deterioration caused by recording or as an estimated repeatability bound. With one retained run per comparison condition, the justified conclusion is that the alternative combination performed worse in these tests, not that the selected gains are globally optimal.

D. Commands and visual evidence

The retained rotor commands are finite. Across the complete final CSV they range from approximately −160.33 rad/s to 162.33 rad/s, including the transient [3]. These are recorded *commands*, not measured motor speeds. Their negative values are consistent with the simulator's signed allocation rule in Equation (4); they are not evidence that a real vehicle should reverse its motors. The full and late-window command histories are shown in Figure 5.

Figure 6 identifies the official TESSE source and shows the desired and actual paths. The video and telemetry complement one another: the recording shows motion in the running system, while Equation (8) verifies the observed angular progress. Earlier independent rigid-body tests are kept separate. Their much smaller errors cannot be substituted for the official-simulator result, because that plant lacks the same Unity contacts and motor dynamics [3].

VI. PERSONAL DISCUSSION AND REFLECTION

A. A convincing picture can still conceal the main error

My most important conclusion is that the evaluation should be designed around the task, not around the most attractive view. In a plan view the achieved circle looks close to the reference, and the radial RMS error is only 0.0829 m. However, the mean height deficit is about 0.253 m. The vertical component alone accounts for 72.1% of the squared position error. Calling this simply "accurate tracking" would overlook the direction in which the system performs least well.

For my validation role, this changes the purpose of the figures. The plan view answers whether the vehicle follows the circular locus; the side view answers whether it holds the requested altitude; the time-domain error answers whether it is near the correct point at the correct time. None replaces the others. I would retain all three forms of evidence even when one of them makes the result appear less favourable.

B. The evaluation window is part of the result

The contrast between the whole-record RMS of 3.1934 m and the late-window RMS of 0.3063 m is not a contradiction. They answer different questions. The former includes acquisition from an offset initial condition; the latter describes the later repeated motion. My interpretation is that a useful report should disclose both, rather than implying that the late-window accuracy applies immediately after launch.

Using two nominal periods is a reproducible convention inherited from the existing analysis, but it is not evidence that the transient must always end after exactly two periods. For a future study, I would specify both a fixed reporting interval and a separate settling criterion before running the comparisons. I would also repeat each gain combination from matched initial states. Those repetitions are not present here, so confidence intervals and claims about reliability across runs would be unsupported.

C. Time alignment deserves the same care as the control equations

The 1,832 repeated reference timestamps are a useful warning against interpreting a nominal publish rate as the actual information rate. New vehicle states can be recorded while the monitor still holds an old reference. Consequently, the current RMS measures error against the reference available to that monitor, not an exactly synchronised continuous-time reference.

I would not remove repeated-reference rows merely to make the data look cleaner: doing so would discard valid state observations and change the metric. A better future logging design would retain source timestamps and receive times for both messages, define a shared time origin, and compare timestamp-aligned states and references. An analytical-reference comparison could then be reported *alongside* the existing latest-reference metric. The present CSV does not independently resolve all communication delays, so I do not claim that asynchrony explains a particular fraction of the observed error.

D. My next experiment would target a cause, not just a number

The gain comparison suggests that increasing several feedback terms is not enough to improve this system. Moreover, the alternative also reduces angular-rate damping, so it does not establish the isolated effect of higher stiffness. My next test would vary one factor at a time while keeping the initial state, trajectory and reporting window fixed.

I would prioritise the vertical channel because its error contribution is largest and a deficit is already visible in the

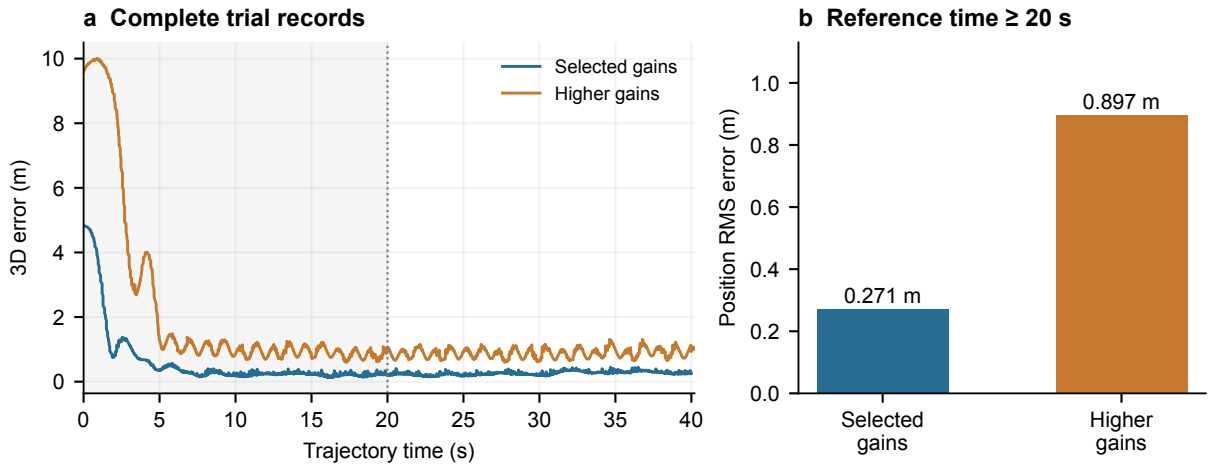


Figure 4. Two official-simulator circular trials. Both summaries use reference time $t \geq 20\,\mathrm{s}$. "Higher gains" is the archive's label for the alternative combination: k_x, k_v and k_R increase, but k_Ω decreases. The experiment therefore compares parameter combinations, not the isolated effect of a single gain. *Source: Group 6 experimental archive.*

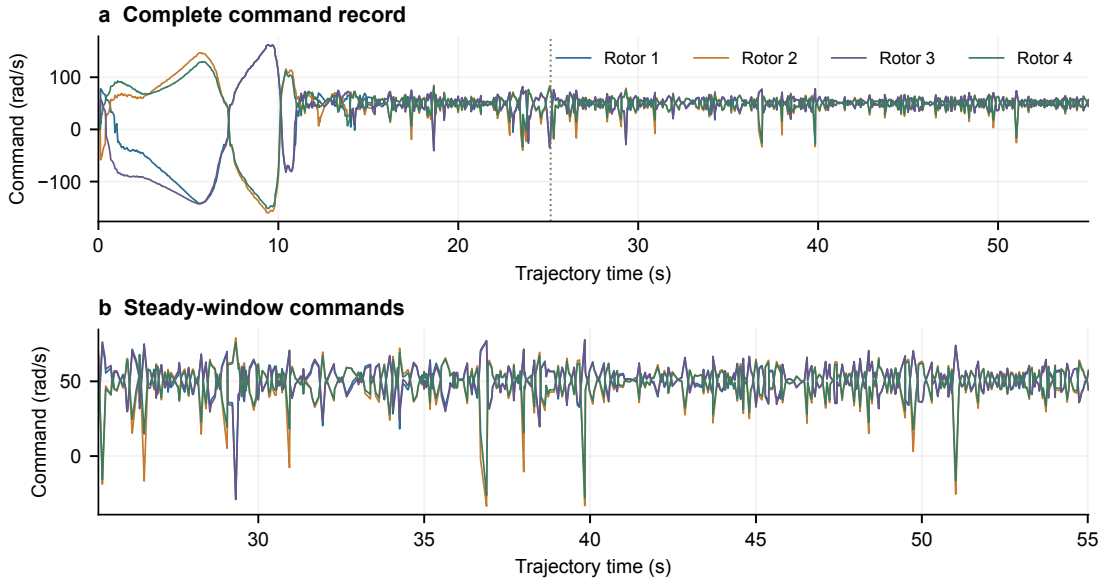


Figure 5. Official TESSE run: the four commanded signed rotor angular velocities and their late-window variation. These are commanded values, not measured motor speeds. *Source: Group 6 experimental archive.*

hover record. A useful sequence would be a repeated static-height test, inspection of the thrust and timing assumptions, and only then a controlled change to the altitude feedback. A bounded integral term with anti-windup is one possible extension, but it would need explicit actuator limits and fresh validation; it is not part of the present controller. I would also investigate a smooth transition into the moving circle instead of starting immediately with a 2.5 m/s reference. These are proposed experiments, not results already demonstrated by the archived runs.

VII. Conclusion

The joint implementation completes repeated circular motion in the official TESSE simulator. My contribution in this report is to connect the analytical reference with measurable acceptance evidence. The 55.02 s final record contains 2.376 observed turns in the documented final window, with a position RMS error of 0.3063 m and a maximum of 0.5845 m. The persistent height deficit and the much larger start-up error limit the interpretation of that success. The results support repeated circular tracking under the recorded conditions, not zero-error tracking, guaranteed real-time control or a universal stability claim.

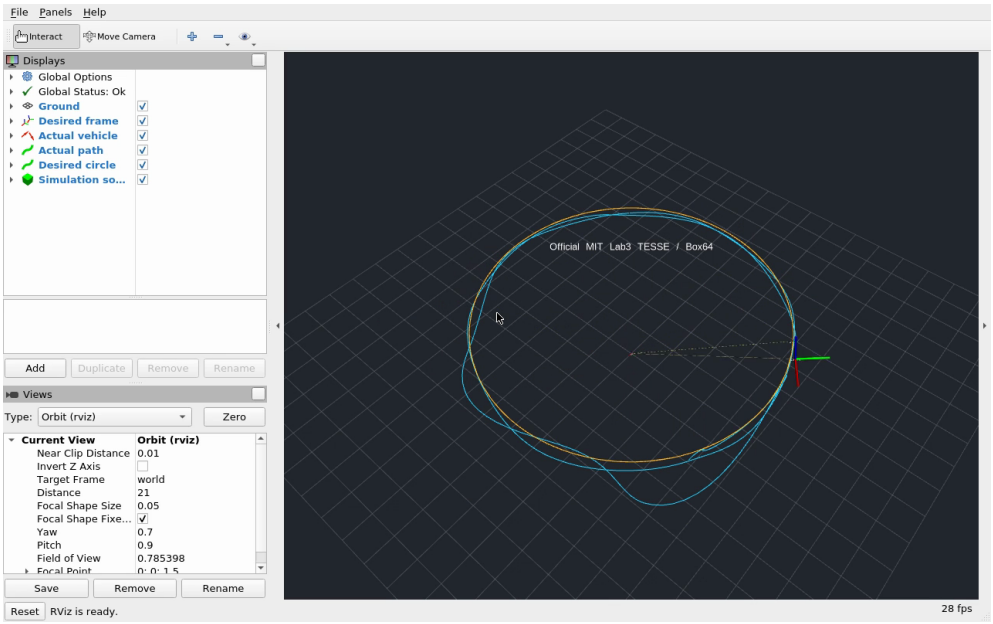


Figure 6. Frame from the official TESSE RViz recording. The image is an original experimental screenshot from the supplied archive, not a rendered illustration or an independent-model demonstration. It provides spatial context for the numeric results, but a single frame does not establish timing accuracy or completion of a full lap.

For Role D, the main lesson is that reference timing, evaluation windows and figure selection are part of the engineering method. A report is more useful when it makes those choices visible and distinguishes measured results from the next improvements to be tested.

VIII. Reproducibility and Evidence

The analysis is based on the supplied Group 6 archive and the shared implementation, not on newly claimed flights. The included `data/unity_flight.csv`, `data/unity_metrics.json` and `data/trials/` document the official TESSE results. The script `analysis/recompute_metrics.py` independently recalculates tracking statistics using the supplied CSV records, without ROS or Python packages beyond the standard library. The archive's `data/unity_bag_info.txt` lists ROS topic counts; the original bag is not present in this package. Figures are included locally in vector PDF wherever available, and the RViz image is a frame from the official TESSE recording. The auxiliary independent rigid-body model is not treated as an official-simulator test. This report does not claim that an external public video link has been submitted. Individual written Deliverables 1 and 2 are independent of this optional ICRA-style course report.

References

[1] MIT VNAV. *Lab 3: Exercises*. 2023. Course task specification. https://vnav.mit.edu/labs_2023/lab3/exercises.html.

[2] Team Fullcans (Group 6), "Lab 3 geometric controller and trajectory publisher," public course repository, revision 3d8f121, 2026. Online: https://github.com/chixin114514/2026Fall_TeamFullcans.

[3] Team Fullcans, Group 6. *Lab 3 experiment archive*, supplied as `report.zip`. Primary records: `unity_flight.csv`, `unity_metrics.json`, `trials/`, `validation.md`, `unity_bag_info.txt`, `numerical_validation.txt`, and `video/rviz_official_frame.png`. Retained shared experimental figures in `lab3_report/figures/`. Report analysis date: 16 October 2026.

[4] T. Lee, M. Leok and N. H. McClamroch. "Geometric tracking control of a quadrotor UAV on SE(3)." *49th IEEE Conference on Decision and Control*, 2010, pp. 5420–5425. doi:10.1109/CDC.2010.5717652.

[5] ROS geometry2 contributors. *tf2_geometry_msgs and tf2 yaw utilities*, ROS Noetic source cited in the supplied exercise answers. https://github.com/ros/geometry2/tree/noetic-devel/tf2_geometry_msgs. https://github.com/ros/geometry2/blob/noetic-devel/tf2/include/tf2/utils.h.

[6] ROS geometry2 contributors. *tf2_eigen quaternion conversions*, ROS Noetic source cited in the supplied exercise answers. https://github.com/ros/geometry2/blob/noetic-devel/tf2_eigen/include/tf2_eigen/tf2_eigen.h.